

Part 1: Conceptual Database Design

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1 Entity-Relationship (ER) Diagram

The conceptual database design was created using the standard Entity-Relationship (ER) notation described in the lectures. The diagram below models all entity sets, relationship sets, key constraints, participation constraints, and class hierarchies (Is-A relationships) as specified in the project requirements.

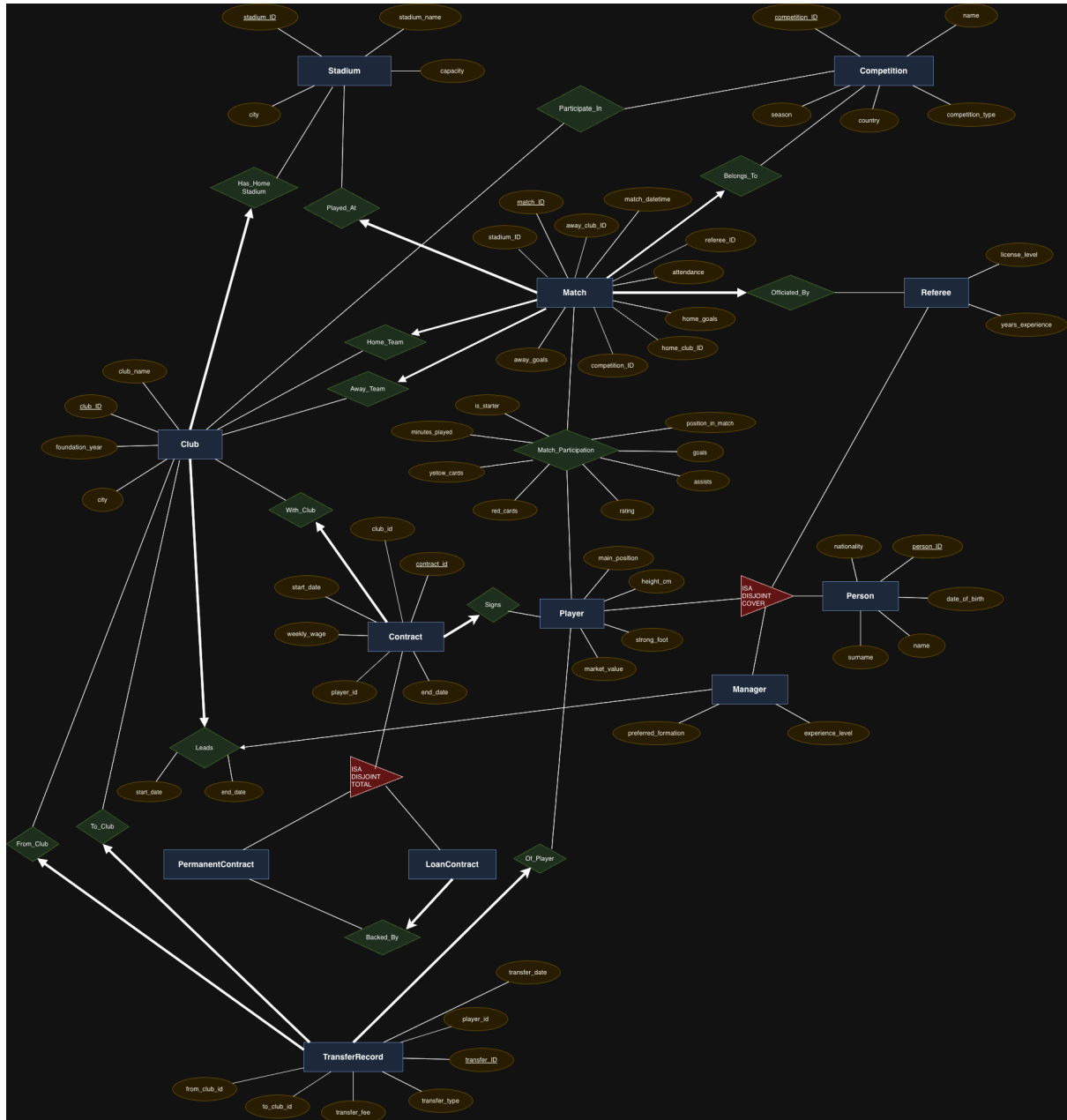


Figure 1: Conceptual ER Diagram for TransferDB

2 Legend

The graphical notation used in the ER diagram follows these conventions:

- **Blue rectangle:** Entity
- **Yellow oval:** Attribute
- **Green diamond:** Relationship
- **Red triangle:** ISA relationship
- **Underlined:** Primary Key
- **Bold arrow toward relation:** Total participation + Key constraint
- **Thin arrow toward relation:** Partial participation + Key constraint
- **Bold line toward relation:** Total participation + No key constraint
- **Thin line toward relation:** Partial participation + No key constraint

3 Constraints Not Expressible in ER Design

While the ER diagram effectively captures the core entities and overall structural relationships, certain complex business rules, cross-table temporal validations, and numeric derivations cannot be natively expressed in standard ER diagramming notation. These constraints will be addressed at the logical layer via SQL `CHECK` constraints or database triggers (or application-level logic). The un-expressible constraints are as follows:

1. (`stadium_name`, `city`) must be `UNIQUE`.
2. (`competition_name`, `season`) must be `UNIQUE`.
3. `attendance` $\leq$ `stadium capacity`.
4. No two matches within 120 minutes at the same stadium.
5. No club in two matches within 120 minutes.
6. No referee in two matches within 120 minutes.
7. Max 11 starters per club per match.
8. Max 23 squad players per match per club.
9. Red/yellow card suspension rules (e.g., 5 yellows = 1 match suspension).
10. Numeric constraints such as `market_value` > 0 , `height_cm` > 0 , and `years_experience` ≥ 0 must hold.
11. The foundation year of a club must be a valid past year.
12. A player may not have two active contracts of the same type at the same time.
13. A player may be associated with at most two clubs simultaneously (one permanent and one loan).
14. The end date of a contract must be strictly later than its start date.
15. Transfer constraints: Free transfer fee = 0; Purchase/Loan fee > 0 ; Loan transfer requires an active Permanent contract with parent club (and the parent club must match `from_club_id`).
16. Manager/Club temporal exclusivity in Leads (a Manager cannot lead two Clubs simultaneously, and a Club cannot have two Managers simultaneously).
17. A person's `date_of_birth` must be strictly in the past.
18. A player cannot appear in the same match as both a home-team and away-team participant.
19. 2 yellow cards in a single match automatically result in a red card (derived state).
20. The Match Result (Win/Loss/Draw) is dynamically derived from home/away goals, rather than explicitly stored as an independent attribute.